Project Title	**DataTalk: Conversational Data Analysis & Auto-Visualization System**
Skills take away From This Project	Python, Streamlit, Pandas, Data Visualization, NLP, LLM Integration, Data Cleaning, Exploratory Data Analysis, Prompt Engineering
Domain	Analytics, Dashboarding, Data Science, Business Intelligence, Automation

Problem Statement

Many businesses struggle to understand and analyze their data because they lack technical expertise in Python, SQL, or BI platforms. Stakeholders need an easy-to-use system where they can:

- Upload their datasets
- Ask natural-language questions
- Instantly get insights
- Automatically generate visualizations

DataTalk solves this problem by providing a conversational analytics engine integrated with a smooth UI for data upload, analysis, and visualization.
The goal is to build a system that transforms uploaded datasets into meaningful insights through intelligent automation and interactive chat.

Business Use Cases

1. Non-Technical Stakeholders
Managers can upload any business dataset and instantly ask: - "Show me monthly revenue trend" - "Which region performed best?" - "Is there a sales drop anywhere?

2. Analysts & Data Teams

- Skip repetitive EDA tasks
- Automatically summarize datasets
- Generate charts quickly

3. Decision Making & Reporting

- Use automated visualizations in presentations
- Quickly answer ad-hoc business questions

4. Learning and Education
   - Students can interactively explore datasets
   - Understand data patterns visually and conversationally

Approach

1. Data Upload

   - User uploads CSV/XLSX files
   - System displays preview, datatype, and summary
   - Supports single datasets

2. Data Preparation

   - Missing value handling
   - Data type conversion
   - Outlier detection using IQR
   - Auto-detect date columns
   - Standardization of column names

3. Exploratory Data Analysis (EDA)

   - The system generates: - Summary statistics - Missing values report - Categorical and numerical analysis -
   - Correlation matrix - Distribution plots
   - A dashboard section will display: - Correlation heatmap - Column-wise distributions - Descriptive statistics

4. Chat With Data (LLM-Powered)

   - The conversational engine: - Understands the user's question - Converts natural-language queries to
   - Pandas operations - Executes code safely - Returns answers with explanations - Automatically generates the
   - best chart for the question
   - Examples: - "Plot the sales vs profit" → scatter plot - "Show the distribution of age" → histogram - "Average price per category" → bar chart

5. Auto-Visualization Engine

- The system automatically selects the best chart type: - Bar chart - Line chart - Scatter plot - Histogram -Boxplot - Pie chart
- Chart selection depends on: - Column type - Data cardinality - Intent of the user question

6. Streamlit Application

- The Streamlit UI will include:
- 📂 **Data Upload Page**
  - Upload CSV/XLSX
  - Preview dataset
  - Display metadata
- 📊 **EDA Dashboard**
  - Summary statistics
  - Correlation heatmap
  - Missing values report
  - Outlier detection
  - Ask any question
  - System returns insights and graphs and Interactive chat window
- 📈 **Auto Visualization Page**
  - User chooses chart type
  - Select x-axis, y-axis
  - Graph rendered instantly
- ⚙️ **Settings Page**
  - Model selection (optional)
  - Theme customization [Auto]
  - Data cleaning preferences

Deployment

- The application can be deployed on: - Streamlit Cloud - Render - Railway - AWS / GCP / Azure - GitHub Pages
- (Streamlit)
- Architecture:
  - 1. Streamlit UI
  - 2. Backend Python Logic
  - 3. LLM for Question Interpretation
  - 4. Pandas for Data Computation
  - 5. Visualization Engine (Matplotlib/Plotly)

Dataset

- This project supports any dataset uploaded by users.
- Example datasets: - Sales / Revenue - HR employee data - Finance data - Retail transactions - Customer
- profiles - E-commerce logs - Healthcare datasets - Supply chain data

Types of Questions the System Can Answer

Category	User Query / Request Examples
1. Descriptive Statistics	"What is the average price?"• "Show the top 10 rows sorted by revenue."
2. Trends & Time Analysis	"Month-wise sales trend"• "Year-over-year comparison"
3. Comparisons	"Which region performed best?"• "Compare revenue across categories."
4. Statistical Insights	Correlation matrix• Outlier detection• Standard deviation analysis
5. Data Quality Checks	"Missing values per column"• "Columns with outliers"
6. Visualization Requests	"Plot sales vs profit"• "Bar chart for quantity per category"• "Distribution of age"

Results (What the App Will Deliver)

The Streamlit application will include:

- ✔ Data upload & preview
- ✔ Automated EDA dashboard
- ✔ Chat-based data insights
- ✔ Auto-generated graphs

- ✔ Downloadable charts
- ✔ Clean and intuitive UI
- ✔ Seamless LLM-powered interaction

Application Sections

The app will have the following sections:

- 🏠 Introduction
- 📂 Upload Dataset
- 📊 EDA Dashboard
- 💬 Chat with Data
- 📈 Auto Visualizations
- 👩‍💻 About Creator

Project Evaluation Metrics

The project will be evaluated based on:

- Accuracy of insights
- Quality of visualizations
- LLM question interpretation accuracy
- Performance on large datasets
- User experience and interface design
- Code readability and structure

Technical Tags

Technologies and skills involved:

- Python
- Pandas

- Streamlit
- LLM
- NLP
- Data Visualization
- Data Cleaning
- EDA
- Prompt Engineering

Deliverables

The project must include:

- Data preparation scripts
- Streamlit app code
- LLM integration logic
- Auto-visualization engine
- Documentation & README
- GitHub repository with code
- Deployment URL

References:

Project Live Evaluation Metrics	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit recordings (English)	Special session for STREAMLIT(11/...

Streamlit recordings (Tamil)	Streamlit - Tamil
Streamlit documentation	Install Streamlit str_app.py Streamlit App on Streamlit Cloud
Project Excellence Series[Deep learning] (Tamil)	Project Excellence Series: Guided Le...
Project Excellence Series[Deep learning] (English)	Project Excellence Series: Guided Le...
Project Excellence Series[NLP] (Tamil)	Project Excellence Series: Guided Le...
Project Excellence Series[NLP] (English)	Project Excellence Series: Guided Le...

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (3:30PM to 4:30PM)

Booking link :https://forms.gle/XC553oSbMJ2Gcfug9

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:
For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

Evaluation Metrics : Project Live Evaluation